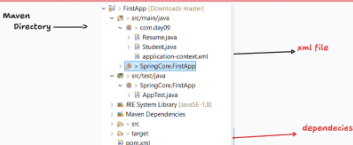


[- Setter and constructor Injection



```
package com.day09;

/*
 * CONSTRUCTOR
 * SETTER
 */
public class Student {
    String name;
    int id;
    String address;
    Resume resume;

    public Student() {
        System.out.println("Student class constructor called ");
    }

    public Student(String name, int id, String address, Resume resume) {
        super();
        System.out.println("parameterized constructor called");
        this.name = name;
        this.id = id;
        this.address = address;
        this.resume = resume;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        System.out.println("name setter called");
        this.name = name;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        System.out.println("Id setter called");
        this.id = id;
    }

    public String getAddress() {
        return address;
    }

    public void setAddress(String address) {
        System.out.println("ADDRESS SETTER CALLED");
        this.address = address;
    }

    public Resume getResume() {
        return resume;
    }

    public void setResume(Resume resume) {
        System.out.println("Resume SETTER CALLED");
        this.resume = resume;
    }

    @Override
    public String toString() {
        return "Student [name=" + name + ", id=" + id + ", address=" + address + ", resume=" + resume + "]";
    }
}
```

```
package com.day09;

public class Resume {
    public String degree;
    public int experience;
    public Resume() {
        System.out.println("Resume class constructor called");
    }

    public Resume(String degree, int experience) {
        super();
        System.out.println("Resume class parameterized");
        this.degree = degree;
        this.experience = experience;
    }

    public String getDegree() {
        return degree;
    }

    public void setDegree(String degree) {
        System.out.println("Resume degree setter called");
        this.degree = degree;
    }

    public int getExperience() {
        return experience;
    }

    public void setExperience(int experience) {
        System.out.println("Resume experience setter called");
        this.experience = experience;
    }

    @Override
    public String toString() {
        return "Resume [degree=" + degree + ", experience=" + experience + "]";
    }
}
```

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:context="http://www.springframework.org/schema/context"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        http://www.springframework.org/schema/beans/spring-beans.xsd
        http://www.springframework.org/schema/context
        http://www.springframework.org/schema/context/spring-context.xsd">

    <bean id="studentResume" class="com.day09.Resume">
        <!-- <property name="degree" value="B.tech"></property>
        <property name="experience" value="1"></property> -->

        <constructor-arg value="B.tech"></constructor-arg>
        <constructor-arg value="1"></constructor-arg>
    </bean>

    <bean id="studentData" class="com.day09.Student">
        <!-- <property name="name" value="prashant"></property>
        <property name="id" value="101"></property>
        <property name="address" value="hyd"></property>
        <property name="resume" ref="studentResume"></property> -->

        <constructor-arg value="prashant"></constructor-arg>
        <constructor-arg value="101"></constructor-arg>
        <constructor-arg value="hyd"></constructor-arg>
        <constructor-arg ref="studentResume"></constructor-arg>

    </bean>

</beans>
```

```
public class App {
    public static void main(String[] args) {

        /* Traditional approach
        * System.out.println("\u001B[31m" + "with java approach");
        * Resume resume = new Resume();
        * resume.setDegree("B.tech");
        * resume.setExperience(0);
        *
        * Student student1 = new Student();
        * student1.setId(101);
        * student1.setAddress("hyd");
        * student1.setName("prashant");
        * student1.setResume(resume);
        *
        * System.out.println(student1);
        */

        System.out.println("\u001B[32m" + "with spring approach" + "\u001B[0m");
        ApplicationContext context = new ClassPathXmlApplicationContext("com/day09/application-context.xml");
        /*
        * If the value is injected by the IOC container through setter then that
        * approach is known as setter Injection
        *
        * setter -> property tag
        *
        * and setter will only work when the class object is created
        *
        *
        * If the value is injected by the IOC container through Constructor then that
        * approach is known as constructor injection
        *
        * Constructor -> constructor-arg
        *
        * constructor will execute when IOC CONTAINER CREATE
        */
        Student bean = context.getBean("studentData", Student.class);
        System.out.println(bean);
    }
}
```